SOPRANO SAXOPHONE AND iPSi - TOWARDS A REFINED PERFORMANCE PRACTICE

James Savage
Faculty of VCA and MCM
University of Melbourne
jsava@student.unimelb.edu.au

ABSTRACT

This paper explores the downstream process of developing the techniques and repertoire required to carry a new extended instrument, the soprano saxophone and iPSi, beyond its initial "novel" status and towards the goal of a truly refined performance practice.
The purpose of iPSi is to facilitate chromatic control of a polyphonic pitch shifting plugin to produce real-time harmony with a single-line instrument. Challenges around setup and the reinforcement of acoustic and processed, amplified sound in a performance or rehearsal environment are addressed with specific reference to the soprano saxophone, although solutions could equally be applied to other instruments in the woodwind family. The paper also traces the development of additional mechanisms of control and refinement of the physical hardware of the interface.

1. INTRODUCTION

New examples of hybrid, meta-, hyper- and extended instruments based around acoustic brass and woodwind instruments surface each year at conferences such as NIME and ICMC. A feature demonstration and discussion of potential musical applications is often given, but there is little evidence that many of these instruments subsequently go on to be fully explored through new bodies of musical work.

In 2012 I designed and built the *harmonic table pitch shifter*, now known as the *isomorphic pitch-shifting interface* and referred to as *iPSi*. A seated instrumental performer's feet play the isomorphic keyboard surface. When the buttons, representing chromatic intervals over a transposable 2.5 octave range are pressed, MIDI output is produced to control pitch-shifting software. The shifted pitches blend with the live instrumental performance to produce harmonies in real time.

Examples of writing on the process of developing material for a new extended or hybrid instrument are rare in comparison to technical papers regarding their development, construction and features. The overarching purpose of my practice-based research is to report, from the perspective of an improvising performer/developer, on the downstream process of taking my extended instrument, the soprano saxophone and iPSi, beyond 'novel' status and developing refined pieces for solo performance. This paper discusses approaches taken to refine the electronics, control and sound production of a new extended instrument to address issues emerging during exploration of the instrument and development of early repertoire.

1.1. Brass, Woodwind and Electronics

Brass and woodwind practitioners have established several distinct approaches to extending their sound palette with technology, the first being the use of conventional effects processors or digital audio plugins to loop and process sound or to add harmonies. Jon Hassell employs fixed-voicing pitch shifting (Hassell 2009) to add single intervals such as perfect fifths or triadic voicings which move in parallel with his live playing, while Arve Henriksen (Henriksen 2008) and Peter Knight (Knight 2012) apply less linear approaches; sampling, looping or applying granular processing to their instrument to create harmonic or textural backgrounds for their melodic playing. Håkon Kornstad creates functional harmony in real time using impressive control over multiphonics and split tones on tenor and bass saxophones; these harmonies are layered and added to with a hardware looper (Kornstad 2009).

Augmented or hybridised acoustic instruments represent a more integrated approach, where the instrument itself is modified to include an array of sensors, switches, potentiometers and other devices in the pursuit of additional avenues of expressivity or intuitive means to control digital audio processing parameters. Hans Leeuw's *Electrumpet* (Leeuw 2009) and Sébastien and Traube's *Electronically-Augmented Saxophone* (2006) follow this train of thought, with both instruments including a means of tracking performer gestures via accelerometer control. Matthew Burtner's *Metasaxophone* (2002) and *SABRe* (Scheisser & Schacher 2012) deeply couple the mechanisms of the acoustic instrument and electronics; the former incorporating force-sensitive resistors on every saxophone key touch, the latter utilising Hall effect sensors to detect the position of bass clarinet keys. Both approaches allow expressive continuous control data to

be derived from key states that were previously only "open" or "closed".

Further innovators discard the conventional acoustic instrument altogether and rebuild from the ground up to create still more radical interfaces for electronically enhanced performance. Melbourne's "Bent Leather Band" perform with several such instruments, including the Serpentine-bassoon and Contra-monster (Favilla, Stuart, and Cannon 2006). They create otherworldly free music sonics using a mixture of processed acoustic sound and live 3D spatialisation.

A common element between the three above-mentioned approaches is the individuality and specificity of each design. The extended instruments are usually designed and implemented by the performers themselves, and each setup features a bespoke feature set catering to the desires and tastes of the individual. Systems are rarely transferrable from one instrument to another, for example it is not a simple process to transpose the mechanisms found on the Electrumpet in order to make them suitable for a saxophone-based system. Each instrument or system delivers unique opportunities for musical expression which may be explored and exploited through the development of a refined performance practice and body of musical works.

1.2. The MIDI Footpedal and iPSi

The MIDI footpedal represents a useful interface for live control of audio processing software or hardware for instrumental performers whose hands are tied up in the very action of playing their instrument. More often than not, MIDI footpedals are designed for switching of presets and manipulation of a limited number of continuous control parameters, but in themselves do not represent intuitive devices for actually "playing" musical material. For example, trumpeter and laptop artist Peter Knight often manipulates a pair of foot treadles to control X/Y plugin parameters (Knight 2012). Organ pedal MIDI foot controllers featuring a linear keyboard scale could be more conducive to musical playing; however, the pitch range is limited and there is little scope to produce more than one or two notes at a given time.

Peter Davies' Opal[1] and the Axis 49 by C-thru Music[2] are both hand-played MIDI interfaces that leverage hexagonal isomorphic layouts to provide an easily navigable, musically intuitive key array with benefits such as single-digit triads, transpositionally invariant scale/chord forms and closely-located consonant pitches such as perfect fifths and major or minor thirds. By applying a similar layout to a MIDI footpedal prototype, I created a MIDI footpedal that was better suited to controlling MIDI tone engines and, more importantly, parameters for live digital harmonisation of instrumental performance (Savage 2013).

1.3. Precipitation of Additional Principles

In his framework for the evaluation of digital musical instruments, O'Modhrain (2011 p. 32) defines principles within digital music instrument design as representing "general design goals, expressed in terms which provide little explicit direction as to how they should be implemented". When I first began designing an interface for real-time harmonisation I had certain top-level principles in mind. The interface would allow a single-line instrumentalist to produce chromatic and diatonic harmonies with up to four voices. The device would not have a static pitch reference, but would instead transpose by fixed chromatic intervals with respect to to the pitch of the live instrumental performance. Harmonised voices would have an ADSR envelope which could be adjusted to facilitate a range of musical contexts.

As development continued and solutions for these core challenges were found, certain lower-level principles also became apparent, biased by my lived experience using technology in music performance and pragmatic concerns that emerged as the extended instrument was put into practice. Concerns around the amount of equipment involved, time taken to setup and repeatability of the setup arose. Ergonomics in practice and performance and the ability to communicate on stage while playing in a physically expressive manner were factors that were initially easy to overlook, but in fact would be crucial to developing a refined performance practice. Compatibility with different computer systems became desirable, as did a simplified and straightforward build for future iterations of the interface.

Thumbswitches placed on the acoustic instrument itself had opened up new musical and technical options such as triggering audio *freezing* for sustained accompaniment and a *damper* function to facilitate smooth footpedal transitions. Expanding upon these simple controls could yield additional functionality, but the drive to explore this avenue was tempered by a desire to minimise impact on the acoustic instrument itself and to keep to a protocol that would allow creation or adaptation of controls to other single-line instruments.

In summary, the new low-level principles were: minimal equipment requirements, ease and repeatability of setup, improved ergonomics and freedom of movement, compatibility across computer systems, simplified physical build and additional, adaptable controls.

Section 2.1.1 contains information on a simplified approach to the wiring scheme for the interface and the transition to presenting the device as a class-compliant MIDI interface for enhanced compatibility across

[1] "History | the Shape of Music | Opal Keyboards," Peter Davies, http://www.theshapeofmusic.com/overview.php.
[2] "C-Thru Music Miscellaneous," C-Thru Music, http://www.c-thru-music.com/cgi/?page=info_misc.

various computer systems and audio plugins. The development of a replacement saxophone thumbrest with additional electronic controls is also discussed. Section 2.1.2 describes how the approach taken to improve amplified sound quality lead to improved ergonomics and a lighter equipment loadout.

2. TOWARDS A REFINED PERFORMANCE PRACTICE

Throughout 2012 I developed a number of interface prototypes - then called the harmonic table pitch shifter, *HTPS*. I had composed a limited amount of material for saxophone and HTPS and given a handful of performances. While I was satisfied with the chromatic pitch-shifting harmonisation mechanic and was growing more comfortable with the interface and accompanying mental bandwidth challenge of coordinating hands, feet, breath and balance at once, I did not consider the system to be complete or the musical outcomes refined.

2.1. Technical experimentation

From a technical perspective, there were issues with hardware reliability, and compatibility across different computer systems. I also felt the rudimentary thumbrest-mounted switches I had incorporated could be expanded upon for enhanced control. Finally, miking, monitoring and amplification were variables that could be difficult to control across varying performance venues. A streamlined system that would be more productive in day-to-day use while still offering enhanced options for sound reinforcement was called for.

2.1.1 MIDI Hardware and Control Mechanisms

Early iPSi iterations incorporated a disassembled USB-MIDI interface[1] to transfer MIDI messages from iPSi's Arduino Mega 2560 to the computer system. One concern was that the device required installation of drivers, which might not be updated/supported into the future. The soldered connections to the board were also fiddly and mechanically weak, again a failure risk.

Class-compliant HIDUINO MIDI firmware (Diakopoulos, Dimitri and Kapur 2011) installed on the Arduino's ATmega16U2 presented an elegant solution to this problem, and in testing has proven reliable across a variety of computer systems. A secondary advantage is the possibility of recompiling the firmware from edited source code to assign an appropriate name to the interface for easy identification.

Early prototypes also used custom stripboard PCBs to consolidate pull-down resistors and other components for connection to the 29 buttons. These were eliminated in the third iteration of the interface, which uses a bare minimum of components (see Table 1) and no additional PCBs, instead using Arduino's internal pullup resistors with button logic inverted at the programming stage. This allows an extremely reliable device to be manufactured quickly and for a low cost.

iPSi footpedal interface	
Arduino Mega 2560 R3	1
Sanwa arcade buttons	29
32-way ribbon cable w/ male connector on one end	1
10K "B" rotary potentiometers	8
Female USB "B" jack with male cable (out to Arduino)	1
RJ45 keystone jack w/ short length of Ethernet cable	1
Wire to distribute +5V and ground connections	

Table 1. List of electronic components, latest iPSi build.



Figure 1. Thumbrests; a. lever microswitches only b. RJ45 connector and DIN switches added c. Forza-inspired enlarged rest.

It was desirable to expand the number of available thumb/hand-operated controls, which also required a connecting cable with multiple cores for different signals. For build simplicity direct connections were preferred to solutions involving multiplexing/charlieplexing methods. The saxophone thumb rest is comparatively small, so a small connector was preferable. While DIN connections are sometimes used in these circumstances, connectors and cable are hard to come by and a faulty bespoke cable could make or break a performance. Common RJ45 connectors and cable were ultimately selected, for several attractive qualities: eight cores meant +5V, ground and six control signals could be accommodated, Ethernet patch cables are cheap and easy to come by – plus they have a locking connector and the female receptacle would take up minimal space on a thumbrest-sized PCB. As shown in Figure 1, several thumbrest prototypes were created, the latest based on similar principles to the Ton Kooiman Forza rest for saxophone, but incorporating two rotary potentiometers, two momentary lever microswitches and two DIN toggles. The levers are assigned to audio "freeze" and "damper pedal" functions, while the

[1] Roland UM-1X

potentiometers are mapped to multiple parameters in effects chains and the DIN toggles are mode selectors, not usually operated during active playing.

2.1.2 Microphone techniques and approaches to amplification

During 2012 I explored variations of a typical two-microphone approach for my instrument, the straight soprano saxophone (Savage 2013). One microphone was placed at the bell of the instrument, and another over the keywork in the vicinity of my top hand. The brightness of the bell sound and warmth of the body sound were blended to produce a reasonable representation of the instrument sound in the room. In most cases, small diaphragm condenser microphones were used, although when spill and feedback became an issue in louder environments a dynamic microphone was substituted at the bell. Such spill and feedback was extremely undesirable as unwanted harmonies were created when pitch-shifted material was looped back through the system.

Amplification was initially provided by a 100-watt full range speaker placed behind the playing position, in the manner of a guitar amplifier. I began experimenting with stereo placement of the pitch-shifted signals and moved to a small 2.1 monitoring system designed for digital drums. These speakers were still placed behind the performance position to provide adequate monitoring, but I would setup perpendicular to these speakers - the microphones picked up less of the amplified sound as they were now off-axis with respect to the speakers, as shown in Figure 2.

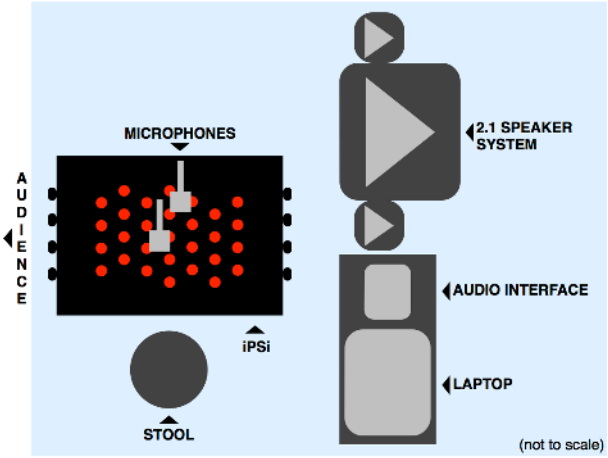


Figure 2. Setup with fixed microphones and 2.1 channel speaker system

The experimentation with microphones and amplification had improved the overall sound and monitoring quality from a performer's perspective but still hadn't eradicated spill and feedback issues. The amount of equipment required had also escalated beyond a point that was practical for day-to-day practice and rehearsals. The combination of the straight soprano saxophone with a fixed two-microphone technique was becoming problematic due to the poor ergonomics of holding this rigid position for long periods of time and the lack of ability to move during performances to communicate with associate performers or simply produce physical expression.

A move to a fully curved soprano saxophone delivered several benefits over the more common straight soprano—an important one being that a single DPA 4099S[1] supercardiod clip-on condenser microphone was effective at capturing the full tone. Freer movement and physical expressivity became possible as a fixed microphone stand was no longer necessary, and I was able to reorientate my performance position to face the audience directly due to the focussed pickup pattern. A woodwind pickup[2] mounted on a spare neck provided a higher level of isolation and this signal could also be used to sidechain gate the clip-on microphone in loud environments, reducing spill and allowing a louder amplified output. There was now sufficient isolation to experiment with highly saturated audio effects on the pitch-shifted signal. Combinations of overdrive and resonant low-pass filtering could be adjusted to blend or contrast with the acoustic saxophone tone and the effects were responsive to live dynamic and tonal variations.

These processed timbres were better reproduced through a guitar amplifier as opposed to a full-range speaker, as the high-frequency tweeters lent an unpleasantly brittle quality to overdriven tones. The lowest note the iPSi system produces with soprano saxophone is Gb_0, which is the instrument's lowest note, Ab_3, transposed down 3 octaves and a major second. While the fundamental frequency of such a note (≅23Hz) sits well outside the optimal range for a guitar amplifier, in practice this unusually low range is not utilised and indeed most playing occurs at pitches of 80Hz and up - above the approximate pitch of the lowest note of a guitar in standard tuning.

These advances in the production of the amplified sound lead to an enhanced feeling of connectedness between the acoustic instrument and the pitch shifted, harmonised output. Placing the amplifier close behind the playing position improved monitoring and could improve intelligibility from an audience perspective, as the acoustic and amplified sounds would come directly from the performance position (see figure 3 below). The pitch shifted signal is necessarily summed to mono before reaching the line-in of the guitar amplifier, but this small compromise has reduced the weight of equipment and setup time dramatically. In a larger performance situation a full stereo signal and dry split of the clip-on microphone signal can still be sent to the

[1] http://www.dpamicrophones.com/en/products.aspx?c=item&category=118&item=24344
[2] http://tjbassoon.com/introducing-the-little-jake-electric-bassoon-pickup/

house PA for sound reinforcement and expanded monitoring possibilities.

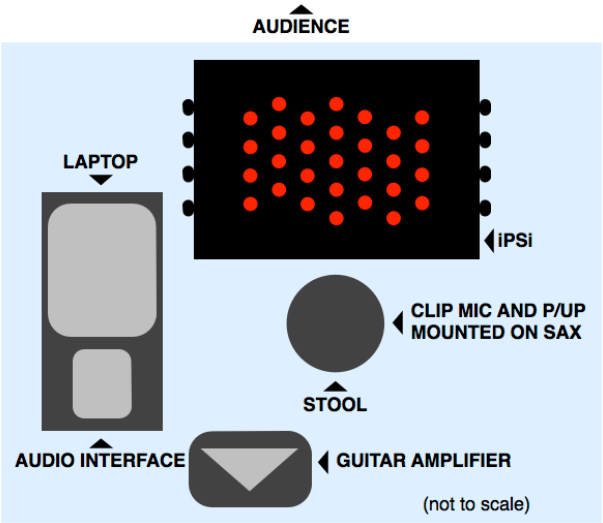


Figure 3. Setup with clip microphone, pickup and guitar amplifier.

2.2. Musical experimentation

An important goal of my practice-based research is the development of refined solo performance pieces for soprano saxophone and the isomorphic pitch shifting interface. Improvisation and the creation of pre-composed material propel the experimental process and cycles of evaluation aim to direct future progress and highlight which approaches deliver successful (and less-successful) musical outcomes.

To date, methods towards developing refined solo performance pieces could be grouped into improvisational, compositional and generative or procedural clusters. Crossover between methods is certainly allowed and there have been cases where a snippet of generative serial material, for example, has been edited and composed and the resultant material used as cells or motifs within an improvisation.

Inspired by a story on John Zorn's approach to composing his prolific "Masada" series, I have lately challenged myself to write a "composition a day" in a small 4-system notebook yielding snippets of 8-12 bars length. This procedure has been carried out in alternating fortnights - one on, one off.

Most commonly, though, video material of improvisation sessions is reviewed and promising motifs or techniques are noted for incorporation into composed segments which may stand alone, or go on to inform or bookend a free improvisation[1]. Therefore improvisation feeds composition and composition feeds improvisation. While this yields a satisfactory-sounding couplet, reality and lived experience came to reveal that such an approach has a tendency towards stagnation – a closed system. The inclusion of generative or procedural processes seeds composition and challenges technique. I am forced to play the footpedal interface in less natural-feeling and sometimes literally uncomfortable ways. One such algorithm was developed to produce serial 2-part inventions based on stochastic variables. As might be expected, the resultant music varies wildly and is sometimes unplayable without alteration, but nevertheless it succeeds in refreshing my concept of the system's capabilities.

[1] Video documentation of certain iPSi performances, pieces and improvisations can be viewed online at http://www.youtube.com/playlist?list=PLvMDtEPaBCihRZb1zv0X81KbaT5wJAEsz

3. CONCLUSION

This paper presents steps towards a refined performance practice for soprano saxophone and iPSi. The process is by no means complete, and further cycles of experimentation and reflection will inevitably lead to further changes to equipment, technique and approach – driving towards the goal of a highly refined extended instrument system and performance pieces exemplifying such a level of refinement.

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